Arduino Timers & Library Conflicts Worksheet

Part 1: Concepts and Definitions

Fill in the blanks or circle the correct answer based on the lecture.

1. To make PWM very efficient and precise without relying on software delays, the Arduino Mega utilizes six built-in hardware timers.
2. **True or False:** If you want to use the Servo library to control a servo, you MUST plug the servo into one of the designated PWM pins.
3. **True or False:** If two different libraries (or a library and an `analogWrite()` command) try to use the same built-in timer at the same time, the Arduino IDE will always give you a clear, meaningful error message before uploading.

Part 2: Pin Conflicts

Using your knowledge of which timers control which pins and libraries, answer the following.

4. The `tone()` function uses **Timer 2** to generate a sound wave. Because of this, which two digital pins can no longer be used for `analogWrite()` (PWM) while `tone()` is active?
   - Pins 9 and 10
5. You are building a robot arm that uses **4 servos**.
   - Which timer does the Servo library automatically take over for the first 1-12 servos? **Timer** 5
   - Because this timer is now in use, you can no longer use `analogWrite()` on which three pins? **Pins** 44, 45, **and** 46
6. You upgrade your robot arm to a massive animatronic project using **16 servos**.
   - The Servo library now needs to use two timers: Timer 5 and Timer 1.
   - Which additional two pins have now lost their native PWM capability? **Pins** 11 **and** 12

Part 3: Troubleshooting Scenarios

Read the following project descriptions. Identify the hardware conflict causing the student's project to fail, and explain why it is happening.

7. **Scenario A:**
Marcus is building a smart nightlight. He has a servo motor plugged into Pin 7 that opens a small door. He also has an LED plugged into Pin 45. He is using `analogWrite(45, 127)` to keep the LED glowing at 50% brightness. However, when he powers on the project, the servo moves, but the LED on Pin 45 acts weird and won't dim properly.

   - **What is the conflict?**

     Servos disable 45-46 / messes up analogWrite/PWM on 45

8. **Scenario B:**
Sarah is building a light/music show. She uses `analogWrite(9, 200)` to control brightness of an LED on Pin 9. She also has a beeper on Pin 3. She writes a code where she turns on the LED, and after a period of time, she uses `tone(3, 440)` to emit a tone. When the tone emits, her LED suddenly stops working correctly.

   - **What is the conflict?**

     Tone uses time 2 → PWM on 9,10 so gets in way of analogWrite on pin 9

Answer Key (For Instructor Reference)

Part 1

1. Timers
2. False (The Servo library generates its signals in software utilizing the timers, so servos can be plugged into any digital pin, not just native PWM pins).
3. PPM; when
4. False (You may or may not see an error; often the code just compiles but behaves erratically).

Part 2

5. Pins 9 and 10.
6. Timer 5; Pins 44, 45, and 46.
7. Timer 1; Pins 11 and 12.

Part 3

8. **Scenario A Conflict:** Marcus is using the Servo library (which automatically hijacks Timer 5 for the first 1-12 servos). Timer 5 controls the native PWM for pins 44, 45, and 46. Because the Servo library is using the timer, Marcus cannot use `analogWrite()` on Pin 45. He needs to move his LED to a different PWM pin governed by a different timer (like Timer 3 or 4).
9. **Scenario B Conflict:** Sarah is using the `tone()` function. `tone()` requires Timer 2 to generate its frequency. Timer 2 is the exact same hardware timer responsible for generating PWM on Pins 9 and 10. Once `tone()` activates, `analogWrite()` on Pin 9 is interrupted. She needs to move her motor control to a different PWM pin not tied to Timer 2.